

Data Analytics -- Guided Notes

Unit tp-5.3 | CompTIA Network+ N10-009 Obj. FC0-U71 5.3 | N10-008 FC0-U71 5.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Data analytics is the process of examining _____ to discover useful information and support decision-making. The four types of analytics in order of complexity are: _____ (what happened), _____ (why it happened), _____ (what will happen), and _____ (what should we do about it).

Question 2

The _____ is the central repository that aggregates data from multiple sources for reporting and analysis. Unlike an operational database (optimized for _____ transactions), a data warehouse is optimized for _____ analytical queries across large historical datasets.

Question 3

ETL stands for _____, _____, and _____. It is the process of moving data from source systems into a data warehouse. _____ pulls data from source systems. _____ cleans and reformats it into a consistent structure. _____ loads it into the destination.

Question 4

Data _____ is the practice of examining large datasets to discover patterns, correlations, and anomalies. It uses techniques from statistics and machine learning. Business applications include _____ (flagging unusual transactions), _____ (anticipating what customers will buy), and _____ (finding patterns that predict equipment failure).

Question 5

_____ is data that is too large, too fast, or too varied for traditional databases to handle. The three Vs of big data are: _____ (massive amounts), _____ (high speed of generation), and _____ (multiple formats and sources). Technologies like Hadoop and Apache Spark process big data across _____ of servers in parallel.

Question 6

A _____ is a visual representation of data designed to communicate patterns and insights clearly. Common chart types: _____ chart for comparing categories, _____ chart for parts of a whole, _____ chart for trends over time, and _____ plot for showing correlation between two variables.

Question 7

_____ are commonly used measures in data analysis. The _____ is the sum divided by the count. The _____ is the middle value when sorted. The _____ is the most frequently occurring value. The _____ is the difference between the maximum and minimum values.

Question 8 -- Real World Application

REAL WORLD: A school cafeteria wants to reduce food waste. The food services director collects 6 months of daily meal sales data. Describe which type of analytics they should use for each of the following goals: (a) understand which meals are most popular, (b) understand why Monday waste is 40% higher than other days, (c) predict how many portions of each meal to prepare next week.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-5-3-data-analytics/tp-5-3-data-analytics-answer-key.pdf
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